Uniform linear upper bound for Berge saturation numbers of every fixed graph

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Editorial note. Complete original mathematical body, all auxiliary proofs, native TikZ figure and embedded bibliography retained. Authors retain responsibility for mathematical correctness; no proof repair, added hypothesis, independent verification or certification.

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Editorial note. Only the main uniform single-graph saturation theorem is counted. Prior partial cases, matching-preserving subset lemma and construction lemma remain complete supporting context/proof, not separate counts.

Allowed standard prerequisites (editorial).

- Berge embeddings, saturation, vertex covers, Hall matchings and degeneracy. Every fixed finite simple F with at least one edge and every fixed k>=2; constants depend only on F,k. Original cited known regimes and their precise bibliographic pointers retained. All new subset-exchange, sparse-construction and remaining-range arguments retained in full.. Author uses known k=2/tree-component/large-vertex-cover cases and proves the remaining range with a matching-preserving indexed-subset construction.

1 Introduction

Let F be a finite simple graph and let $\mathcal{H}$ be a hypergraph with $V(F) \subseteq V(\mathcal{H})$. We say that $\mathcal{H}$ is a *Berge-F* if there is a bijection $\psi : E(F) \longrightarrow E(\mathcal{H})$ such that $e \subseteq \psi(e)$ for every $e \in E(F)$. We call F a *core graph* of $\mathcal{H}$ and its vertices the *core vertices*. For $k \geq 2$, let $\mathcal{B}_k(F)$ denote the family of k-uniform Berge-F hypergraphs, considered up to isomorphism, with isolated vertices outside the core graph omitted. A k-uniform hypergraph *contains a Berge-F* if it has a subhypergraph isomorphic to a member of $\mathcal{B}_k(F)$. The hypergraph is *Berge-F-free* if it contains no Berge-F. It is *Berge-F-saturated* if it is Berge-F-free and adding any missing k-edge creates a Berge-F. For a graph F, the saturation number is

$$\mathrm{sat}_k(n, \text{Berge-}F) = \min\{|E(\mathcal{H})| : |V(\mathcal{H})| = n,\ \mathcal{H} \text{ is } k\text{-uniform and Berge-}F\text{-saturated}\}.$$

For $k = 2$, this is ordinary graph saturation, and we write $\mathrm{sat}(n, F) = \mathrm{sat}_2(n, \text{Berge-}F)$. Its study was initiated by Erdős, Hajnal and Moon [7]. Kászonyi and Tuza [9] proved that $\mathrm{sat}(n, F) = O_F(n)$ for every fixed graph F with at least one edge. For k-uniform hypergraphs, Pikhurko [10] proved an $O(n^{k-1})$ upper bound for every fixed finite family of forbidden hypergraphs. Every member of $\mathcal{B}_k(F)$ has at most $|V(F)| + (k-2)|E(F)|$ vertices, so $\mathcal{B}_k(F)$ is finite. Pikhurko's result therefore gives $\mathrm{sat}_k(n, \text{Berge-}F) = O_{F,k}(n^{k-1})$.

English, Gordon, Graber, Methuku and Sullivan [4] initiated the study of Berge saturation and obtained results for paths, cycles, stars, triangles and matchings. They proposed the following conjecture.

For a family $\mathcal{F}$ of graphs, put $\mathcal{B}_k(\mathcal{F}) = \bigcup_{F\in\mathcal{F}} \mathcal{B}_k(F)$. A hypergraph is Berge-$\mathcal{F}$-free if it is Berge-$F$-free for every $F \in \mathcal{F}$, and is Berge-$\mathcal{F}$-saturated if it is Berge-$\mathcal{F}$-free and adding any missing k-edge creates a Berge-F for some $F \in \mathcal{F}$. We write $\mathrm{sat}_k(n, \text{Berge-}\mathcal{F}) = \mathrm{sat}_k(n, \mathcal{B}_k(\mathcal{F}))$.

Conjecture 1.1 (English et al. [4, Conjecture 6.3])**.** *For every fixed integer $k \geq 2$ and every fixed finite nonempty family $\mathcal{F}$ of graphs, each having at least one edge,*

$$\mathrm{sat}_k(n, \text{Berge-}\mathcal{F}) = O_{\mathcal{F},k}(n).$$

English, Gerbner, Methuku and Tait [5] proved the single-graph case for all $3 \leq k \leq 5$. For arbitrary $k \geq 3$, they established the linear bound when F has a tree component, when its vertex cover number $\beta(F)$ is at least $k+1$, or when it has a component with vertex feedback number at most two. Gerbner, Patkós, Tuza and Vizer [8, Theorems 1.2 and 1.3] subsequently proved linear bounds for complete multipartite graphs with at least three nonempty parts, and for graphs having an edge whose endpoint degrees are the two smallest entries of the degree sequence. English, Kritschgau, Nahvi and Sprangel [6] obtained further linear bounds when $\delta(F) > \beta(F)/2$ or $g(F) > f(F)$, where $\delta(F)$ is the minimum degree, $g(F)$ is the girth and $f(F)$ is the minimum number of vertices whose deletion leaves a forest. Exact and asymptotic saturation numbers, as well as saturation spectra, have also been studied for Berge stars and cliques; see [1, 2, 6, 3].

In this paper, we prove the single-graph case of Conjecture 1.1 for every uniformity.

Theorem 1.2. *For every fixed integer $k \geq 2$ and every fixed finite simple graph F with at least one edge,*

$$\mathrm{sat}_k(n, \text{Berge-}F) = O_{F,k}(n).$$

Our proof builds on the construction of English, Gerbner, Methuku and Tait [5, Construction 2.1] and their argument for bounding the number of additional edges. The case in which we delete two adjacent vertices is also related to the type-I construction of Gerbner, Patkós, Tuza and Vizer [8, proof of Theorem 1.3]. We replace the full family of subsets of a prescribed size by a selected family that admits the required matchings.

The new step is to make this choice uniformly in the remaining range $\beta(F) \leq k$. A degree threshold identifies either two adjacent vertices or an induced star whose vertices can be deleted from F. In the star case, we choose subsets of size strictly less than k. When more than k high-degree vertices remain after deleting the star, a minimum vertex cover gives initial indexed subsets of size at most $\beta(F) - 1$. We enlarge them to $(k-1)$-subsets and make them distinct by an exchange argument, preserving the required matchings. The strict bound below k leaves room for a nonempty part of outside vertices in each hyperedge. A count of high-degree vertices then excludes Berge-F from the construction, and a greedy forest embedding bounds all Berge-F-free extensions.

Section 2 records the known cases and proves the exchange lemma. Section 3 gives the construction, verifies it, and chooses its parameters to prove Theorem 1.2.

2 Preliminaries

All graphs and hypergraphs are finite and simple. A hypergraph is k-uniform if each hyperedge has size k. For a graph F, let $V(F)$ and $E(F)$ denote its vertex and edge sets, and let $N_F(v)$ and $d_F(v)$ denote the neighborhood and degree of v. The degree $d_{\mathcal{H}}(v)$ of a vertex in a hypergraph is the number of hyperedges containing it. For $W \subseteq V(\mathcal{H})$, the induced subhypergraph $\mathcal{H}[W]$ has vertex set W and edge set $\{h \in E(\mathcal{H}) : h \subseteq W\}$. For $R \subseteq V(F)$, let $F[R]$ denote the induced subgraph, and write $F - R = F[V(F) \setminus R]$. For disjoint vertex sets $U, W \subseteq V(F)$, let $e_F(U, W)$ be the number of edges with one endpoint in each set. A *vertex cover* is a set of vertices meeting every edge; its minimum size is the *vertex cover number*, denoted by $\beta(F)$. A forest is an acyclic graph and may have isolated vertices. For a nonnegative integer d, a hypergraph $\mathcal{H}$ is *d-degenerate* if every nonempty induced subhypergraph has a vertex of degree at most d. For a positive integer q, put $[q] = \{1, \ldots, q\}$, and let $\binom{S}{t}$ denote the family of t-subsets of S. For nonnegative integers b, t with $t > b$, we set $\binom{b}{t} = 0$.

We may assume that F has no isolated vertices, since they do not affect Berge saturation for sufficiently large n. Vertices that become isolated in $F-R$ are nevertheless retained in the construction. All constants may depend on F and k, but not on n.

We use the known cases without reproving them. The case $k = 2$ follows from [9]. For $k \geq 3$, graphs having a tree component are covered by [5, Observation 2.4], and the range $\beta(F) \geq k+1$ is covered by [5, Theorem 2.5]. Thus it remains to consider the following range:

$$k \geq 3, \qquad \beta(F) \leq k, \qquad \text{every component of } F \text{ contains a cycle.} \tag{2.1}$$

2.1 Choosing distinct subsets

For an indexed family $P_1, \dots, P_r \subseteq S$, its incidence graph is the bipartite graph with vertex classes S and $[r]$, where z is adjacent to i if and only if $z \in P_i$. Equal sets $P_i = P_j$ still correspond to distinct vertices i and j. The following lemma allows us to make the sets distinct while preserving the matchings needed in the construction.

Lemma 2.1. *Let S be a set of size c, let t be a nonnegative integer, and let $P_1, \dots, P_r \subseteq S$ be indexed sets satisfying*

$$|P_i| \leq t \leq c, \qquad 1 \leq r \leq \binom{c}{t}.$$

There are pairwise distinct sets $P'_1, \dots, P'_r \in \binom{S}{t}$ with the following property: for every $Q \subseteq S$, if the incidence graph of $(P_1, \dots, P_r)$ has a matching saturating Q, then so does the incidence graph of $(P'_1, \dots, P'_r)$.

Proof. First enlarge each P_i to a t-set. This only adds edges to the incidence graph, so every previously matchable subset of S remains matchable. If $t = 0$ or $t = c$, the inequality $r \leq \binom{c}{t}$ gives $r = 1$, and the enlarged family already has the required property. We may therefore assume $0 < t < c$.

We first verify that one elementary replacement preserves all the required matchings. Suppose that two indices $i \neq j$ have the same set $P_i = P_j = P$. Choose $x \in P$ and $y \in S \setminus P$, and replace only P_i by $P_i^* = (P \setminus \{x\}) \cup \{y\}$. Let $Q \subseteq S$ be any set that has a matching in the incidence graph before this replacement, and fix a matching saturating Q. The only incidence removed is the edge joining x to index i. If this edge is not in the matching, the matching remains valid. Otherwise, match x to index j, which still corresponds to P. If j was unmatched, this completes the modification. If j was matched to a vertex z, then $z \neq x$ and $z \in P$; match z to index i instead, since $z \in P_i^*$. The modified matching still saturates Q. Thus this replacement preserves the existence of a matching for every subset Q that was matchable before it.

To eliminate all repetitions using these replacements, consider the Johnson graph $J(c,t)$: its vertices are the t-subsets of S, and two vertices are adjacent if one is obtained from the other by replacing a single element. This graph is connected. Indeed, for two t-sets U and V, we can successively replace an element of $U \setminus V$ by an element of $V \setminus U$ until we reach V.

Place each index $i \in [r]$ at the vertex P_i of $J(c,t)$. If a vertex carries at least two indices, some vertex is unoccupied, because $r \leq \binom{c}{t}$. Fix a vertex carrying at least two indices, and choose an unoccupied vertex at minimum distance from it. Every internal vertex of a shortest path between them is occupied. Move one index along this path, one edge at a time. Before each move its current vertex carries at least two indices, so the replacement above applies. At the end, the number of occupied vertices has increased by one, and all previously matchable subsets remain matchable. Repeat until the r indices occupy distinct vertices. □

3 The construction and proof of Theorem 1.2

We first give the construction and prove that every Berge-F-free hypergraph containing it has only linearly many edges. The bound on the additional edges follows the degeneracy argument in [5, Lemma 2.2]; we include the argument for the more general choice of subsets used here. We then choose the parameters in two cases to prove Theorem 1.2.

3.1 The construction

Construction 3.1. Fix an integer d, sets $A, R, S \subseteq V(F)$, a graph G, integers q, t, and subsets $P_1, \dots, P_q$ satisfying the following conditions:

1. $d \geq 2$,
$$A = \{v \in V(F) : d_F(v) \geq d\}, \qquad R \cap A \neq \varnothing, \qquad S = A \setminus R, \qquad G = F - R.$$

2. The graph $F[R]$ is a forest, and every edge between R and $V(G)$ has its endpoint in $V(G)$ in S.

3. The integers q, t satisfy
$$1 \leq q \leq d-1, \qquad 0 \leq t < k. \tag{3.1}$$

4. The sets $P_1, \dots, P_q$ are pairwise distinct t-subsets of S. For each $v \in R$, put $Q_v = N_F(v) \cap S = N_F(v) \setminus R$. We require an injection
$$\rho_v : Q_v \longrightarrow [q], \qquad z \in P_{\rho_v(z)} \quad (z \in Q_v). \tag{3.2}$$
Equivalently, the incidence graph of the P_j has a matching saturating each Q_v. These matchings are chosen separately for different vertices v.

All the parameters above are fixed independently of n. Put $a = k - t \geq 1$. With these choices, for sufficiently large n, construct a k-uniform hypergraph $\mathcal{H}_0$ on an n-vertex set as follows. Choose $|V(G)|$ vertices and label them by $V(G)$, retaining any isolated vertices of G. For each $xy \in E(G)$, choose a set W_{xy} of $k-2$ additional vertices. These sets are pairwise disjoint and disjoint from $V(G)$. Put
$$h_{xy} = \{x, y\} \cup W_{xy}, \qquad Z_0 = V(G) \cup \bigcup_{xy \in E(G)} W_{xy}.$$

The hyperedges h_{xy} form the k-uniform expansion of G. Partition the remaining vertices into sets $B_1, \dots, B_m$ of size a and a remainder Z_1 of size less than a, where
$$m = \left\lfloor \frac{n - |Z_0|}{a} \right\rfloor, \qquad |Z_0| = |V(G)| + (k-2)|E(G)|.$$

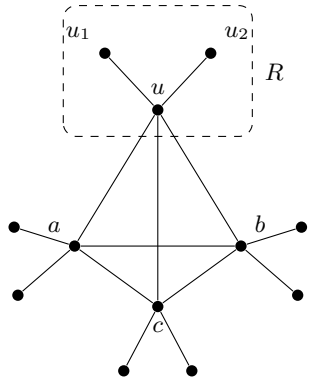
Thus
$$V(\mathcal{H}_0) = Z_0 \,\dot\cup\, B_1 \,\dot\cup\, \cdots \,\dot\cup\, B_m \,\dot\cup\, Z_1, \qquad |B_i| = a, \quad |Z_1| < a.$$

Set $Z = Z_0 \,\dot\cup\, Z_1$ and define the edge set by
$$E(\mathcal{H}_0) = \{h_{xy} : xy \in E(G)\} \;\cup\; \{B_i \cup P_j : i \in [m],\ j \in [q]\}. \tag{3.3}$$

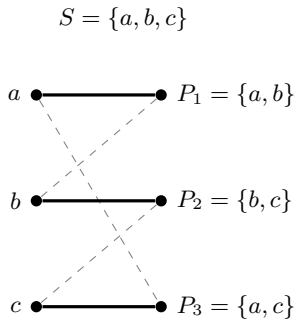
Every hyperedge has size k, and all hyperedges are distinct: the P_j are distinct, every B_i is nonempty, and no h_{xy} meets a part B_i. The vertices of Z_1 are isolated. Since G, q and a are fixed independently of n, we have
$$|E(\mathcal{H}_0)| = |E(G)| + qm = \frac{q}{a} n + O_{F,k}(1) = O_{F,k}(n). \tag{3.4}$$

Moreover,
$$|Z| \leq |V(G)| + (k-2)|E(G)| + a - 1. \tag{3.5}$$

Thus $|Z|$ is bounded independently of n, and the remainder allows all sufficiently large values of n. An example is shown in Figure 1.



(a) The graph F



(b) A matching saturating S

Figure 1: Let F be obtained from K_4 by adding two pendant vertices at each vertex, and let $k = 4$. Take $d = 5$, $R = \{u, u_1, u_2\}$ and $S = \{a, b, c\}$. The displayed sets satisfy (3.2): $Q_u = S$ and $Q_{u_1} = Q_{u_2} = \varnothing$. In the resulting hypergraph $\mathcal{H}_0$, each part B_i has size two and contributes the three edges $B_i \cup \{a, b\}$, $B_i \cup \{b, c\}$ and $B_i \cup \{a, c\}$. The hyperedges h_{xy} form the k-uniform expansion of $F - R$. In (b), lines are edges of the incidence graph, not hyperedges.

The next lemma explains the conditions imposed above. The degree bound in (3.1) makes the initial hypergraph Berge-F-free, while (3.2) provides the hyperedges needed for the edges between R and $V(G)$.

Lemma 3.2. *Let $\mathcal{H}_0$ be the hypergraph defined in Construction 3.1. Then $\mathcal{H}_0$ is Berge-F-free, and every Berge-F-free k-uniform hypergraph on the same vertex set containing $\mathcal{H}_0$ has $O_{F,k}(n)$ edges. Consequently, an inclusion-maximal Berge-F-free extension of $\mathcal{H}_0$ is a Berge-F-saturated hypergraph with $O_{F,k}(n)$ edges.*

Proof. We first count the vertices of degree at least d in $\mathcal{H}_0$. Every vertex of a part B_i has degree $q \le d-1$. Every vertex in a set W_{xy} has degree one, and every vertex of Z_1 has degree zero. If $w \in V(G) \setminus S$, then $d_F(w) < d$ and $d_{\mathcal{H}_0}(w) = d_G(w) \le d_F(w)$. Thus only the vertices in S can have degree at least d. In a Berge-F, the hypergraph degree of each core vertex is at least its degree in F, since its incident graph edges require distinct hyperedges containing that vertex. But $|S| < |A|$, because R meets A. There are therefore too few vertices of degree at least d in $\mathcal{H}_0$ to serve as the core vertices corresponding to A. Hence $\mathcal{H}_0$ is Berge-F-free.

Let $\mathcal{H} \supseteq \mathcal{H}_0$ be any Berge-F-free k-uniform hypergraph on the same vertex set, and let $\mathcal{J}$ have the same vertex set and edge set $E(\mathcal{H}) \setminus E(\mathcal{H}_0)$. Put $T = F[R]$. We show that $\mathcal{J}$ contains no Berge-T whose core vertices lie outside Z in pairwise distinct parts B_i. All vertices of T, including any isolated vertices, are retained as core vertices.

Suppose such a Berge-T exists. For each $xy \in E(T)$, let $\theta(xy)$ be the corresponding hyperedge under its defining bijection. For $v \in R$, let $B(v)$ be the part containing the core vertex corresponding to v. Keep the vertices of G at their fixed labels. Assign hyperedges to the edges of F as follows:

edge of F	hyperedge in $\mathcal{H}$
$xy \in E(G)$	h_{xy}
$xy \in E(T)$	$\theta(xy)$
$vz,\ v \in R,\ z \in Q_v$	$B(v) \cup P_{\rho_v(z)}$.

(3.6)

For a fixed v, the last row uses distinct hyperedges by injectivity of ρ_v. Different vertices of R use different parts $B(v)$, so their assigned hyperedges are also distinct. The first row uses no part B_i, and the second row consists of edges outside $\mathcal{H}_0$. Every assigned hyperedge contains the two core vertices corresponding to its graph edge. The fixed vertices of G and the core vertices chosen outside Z are also distinct. These assignments therefore give a Berge-F in $\mathcal{H}$, a contradiction.

It remains to bound the number of edges of $\mathcal{J}$. Put

$$p = |R|, \qquad b = |Z| + ap, \qquad \gamma = \binom{b}{k-1} + p.$$

We claim that $\mathcal{J}$ is γ-degenerate. Suppose, to the contrary, that a nonempty induced subhypergraph $\mathcal{J}[W]$ has minimum degree greater than γ. Then $|W| > b$, since otherwise simplicity gives $d_{\mathcal{J}[W]}(x) \le \binom{|W|-1}{k-1} \le \binom{b}{k-1} < \gamma$. Root each component of T and embed the components successively, placing every parent before its children. We keep the chosen core vertices outside Z in distinct parts and use a different hyperedge for each graph edge. These restrictions on parts apply only to the core vertices, not to the other vertices of the assigned hyperedges.

To start a component, choose a vertex of W outside Z and the parts already used. There are at most $p-1$ such parts, so fewer than b vertices are excluded; hence a choice exists. This also handles isolated vertices of T. Suppose next that a parent has been placed at x and a child remains to be placed. Let Y be the union of Z and the parts already used. Then $x \in Y$ and $|Y| \le b$. At most $\binom{b}{k-1}$ hyperedges through x lie entirely in Y, and fewer than p hyperedges have already been assigned. Since $d_{\mathcal{J}[W]}(x) > \gamma$, there is an unassigned hyperedge $e \ni x$ with $e \nsubseteq Y$. Choose $y \in e \setminus Y$ for the child and assign e to the corresponding graph edge. Since $e \subseteq W$, the vertex y lies in W and in a new part. Repeating this step constructs the forbidden Berge-T.

Thus $\mathcal{J}$ is γ-degenerate. Successively delete a vertex of degree at most γ from the remaining induced subhypergraph, counting each hyperedge when its first vertex is deleted. Each step counts at most γ hyperedges, so $|E(\mathcal{J})| \le \gamma n$. By (3.5), γ is bounded in terms of F and k. Together with (3.4), this gives $|E(\mathcal{H})| = O_{F,k}(n)$. Finally, add missing edges to $\mathcal{H}_0$ one at a time whenever they do not create a Berge-F. Since only finitely many k-edges are available, this procedure terminates at an inclusion-maximal Berge-F-free extension, which is saturated by definition and satisfies the same bound. □

3.2 Proof of Theorem 1.2

Proof. The cited results in Section 2 allow us to assume (2.1). Choose the degree threshold

$$d = \min_{xy \in E(F)} \max\{d_F(x), d_F(y)\}, \qquad r = d - 1, \tag{3.7}$$

and put

$$L = \{v \in V(F) : d_F(v) < d\}, \qquad A = V(F) \setminus L.$$

Since every component of F contains a cycle, F has no K_2 component, and $d \ge 2$. The set L is independent, since an edge with both endpoints in L would contradict (3.7). Also, $|A| \ge 2$: an independent set together with at most one other vertex cannot contain a cycle.

There is a vertex of degree exactly d. Define

$$\ell = \max\{|N_F(u) \cap L| : d_F(u) = d\}.$$

We choose R and the sets P_j according as $\ell \le 1$ or $\ell \ge 2$, and then apply Construction 3.1 and Lemma 3.2.

Case 1: deleting two adjacent vertices ($\ell \le 1$). If $\ell = 0$, choose an edge uv attaining the minimum in (3.7). Both endpoints have degree d, and all their other neighbors lie in A. If $\ell = 1$, choose a vertex u of degree d with its unique neighbor $v \in L$. All other neighbors of u lie in A, and all other neighbors of v do as well, since L is independent. Thus, in either case,

$$d_F(u) = d \ge d_F(v), \qquad (N_F(u) \cup N_F(v)) \setminus \{u, v\} \subseteq A.$$

Take $R = \{u, v\}$, $S = A \setminus R$ and $G = F - R$. Then $F[R]$ is an edge and every edge from R to $V(G)$ ends in S. Write

$$U = N_F(u) \setminus \{v\}, \qquad W = N_F(v) \setminus \{u\}.$$

We have $U, W \subseteq S$, $|U| = r$ and $|W| \le r$. We will choose r distinct sets of a common size, either one or two, whose incidence graph has matchings saturating both U and W.

If $W \subseteq U$, take the r singleton sets $\{z\}$ for $z \in U$. They have matchings saturating both U and W. Otherwise, extend W to an r-set $W' \subseteq U \cup W$. Enumerate $U = \{u_1, \dots, u_r\}$ and $W' = \{w_1, \dots, w_r\}$, and initially take $P_i = \{u_i, w_i\}$. These indexed sets have size at most two and admit matchings saturating

both U and W'. Since $W \not\subseteq U$, we have $|S| \geq |U \cup W| \geq r+1$, and hence $r \leq \binom{|S|}{2}$. Lemma 2.1, applied with $t = 2$, gives r distinct pairs. Use these pairs as $P_1, \ldots, P_r$. Their incidence graph has a matching saturating U and a matching saturating W', and hence a matching saturating W.

In both subcases, $q = r = d - 1$ and t is one or two. Since $k \geq 3$, conditions (3.1) and (3.2) hold. With $a = k - t$ and $G = F - R$, form $\mathcal{H}_0$ as in Construction 3.1. Lemma 3.2 gives the required saturated hypergraph.

Case 2: deleting an induced star ($\ell \geq 2$). Choose u with $d_F(u) = d$ and $|N_F(u) \cap L| = \ell$, and put

$$R = \{u\} \cup (N_F(u) \cap L), \qquad S = A \setminus \{u\}, \qquad G = F - R. \tag{3.8}$$

Since L is independent, $F[R]$ is an induced star and every edge from R to $V(G)$ ends in S. The set S is nonempty since $|A| \geq 2$. Moreover,

$$|Q_u| = d - \ell \leq r - 1, \qquad |Q_v| = d_F(v) - 1 \leq r - 1 \quad (v \in R \setminus \{u\}). \tag{3.9}$$

Here $d \geq 3$: if $d = 2$, the two lower-degree neighbors of u would both have degree one, making their component a three-vertex path, contrary to (2.1). Thus $r \geq 2$. It remains to find the sets satisfying (3.1) and (3.2). Write $c = |S|$.

Case 2.1: $c \leq k$. If $c = 1$, take $q = 1$, $t = 1$ and $P_1 = S$. The matching condition holds because every Q_v is a subset of S. For $2 \leq c \leq k$, put $q = \min\{r, c\}$, choose distinct $s_1, \ldots, s_q \in S$, and take

$$P_j = S \setminus \{s_j\} \qquad (j \in [q]). \tag{3.10}$$

These sets have common size $t = c - 1 < k$. To verify (3.2), fix Q_v and apply Hall's theorem. Each vertex of Q_v belongs to at least $q - 1 \geq 1$ of the sets P_j. Any subset of Q_v of size at least two meets every P_j, and therefore has all q indices as neighbors. Since $|Q_v| \leq \min\{r - 1, c\} \leq q$, Hall's condition holds.

Case 2.2: $c > k$. Let K be a minimum vertex cover and write $\beta = |K| = \beta(F)$. Since $c > k \geq \beta$, there is a vertex $w \in S \setminus K$. Every neighbor of w belongs to K, so

$$d \leq d_F(w) \leq \beta \leq k, \qquad r \leq k - 1. \tag{3.11}$$

Put $B = K \cap S$ and $X = K \cap R$. The set X is nonempty because K covers the edges of the star $F[R]$. The set B is also nonempty. Indeed, otherwise $K \subseteq L \cup \{u\}$, so every vertex of K has degree at most d. All neighbors of S would lie in K, and $S \cap K = \varnothing$. Hence

$$dc \leq \sum_{s \in S} d_F(s) = e_F(S, K) \leq \sum_{x \in K} d_F(x) \leq d\beta < dc,$$

a contradiction.

We first choose r indexed subsets, allowing repetitions and different sizes. Choose $b_0 \in B$ and $x_0 \in X$. For each $x \in X$, assign distinct colors from $[r]$ to the edges from x to $S \setminus K$. Require color 1 to be unused at x whenever $x = x_0$ or $b_0 \in Q_x$. These assignments can be made independently by (3.9), since there are at most $r - 1$ edges to color at each x. No restriction is imposed on colors at their endpoints in $S \setminus K$. For $j \in [r]$, let $E_j \subseteq S \setminus K$ be the set of endpoints of edges assigned color j, and put

$$P_1 = B \cup E_1, \qquad P_j = (B \setminus \{b_0\}) \cup E_j \quad (2 \leq j \leq r). \tag{3.12}$$

Each $x \in X$ contributes at most one vertex to each E_j, and x_0 contributes none to E_1. Consequently,

$$|P_j| \leq |B| + |X| - 1 \leq \beta - 1 \leq k - 1 \qquad (j \in [r]). \tag{3.13}$$

For $x \in X$, match each vertex of $Q_x \setminus K$ to the color of its edge from x. If $b_0 \in Q_x$, color 1 is unused at x, so match b_0 to index 1. Every other vertex of $Q_x \cap B$ belongs to all P_j and can be matched to an unused index. There are enough indices because $|Q_x| \leq r - 1$. For $v \in R \setminus K$, we have $Q_v \subseteq B$. Match

b_0 to index 1 when needed, and match the other vertices to distinct remaining indices. Thus every Q_v has the required matching in the indexed family (3.12).

Finally, take $t = k - 1$. By (3.11) and $c > k$,

$$r \leq k - 1 < c \leq \binom{c}{k-1}.$$

Lemma 2.1 gives r distinct $(k-1)$-subsets of S preserving every required matching. Use these as $P_1, \ldots, P_r$ and put $q = r$. This proves (3.1) and (3.2).

In both subcases, R meets A, $F[R]$ is a forest, and every edge from R to $V(G)$ ends in S. With the choices of q and t above and $a = k - t$, form $\mathcal{H}_0$ as in Construction 3.1. Lemma 3.2 gives the required saturated hypergraph. This completes the proof of Theorem 1.2. □

Declaration on the use of generative AI

During the preparation of this manuscript, the author used GPT-6 Astra (OpenAI) to explore proof strategies, assist with checking proofs, and refine the exposition. The author reviewed and verified all mathematical arguments, results, and conclusions presented in the manuscript and takes full responsibility for its content.

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